

重积分的应用

立体体积

$$V = \iiint_{\Omega} dx dy dz = \iint_D dx dy \int_{z_1(x,y)}^{z_2(x,y)} dz = \iint_D (z_2(x,y) - z_1(x,y)) dx dy$$

(柱坐标) $V = \iiint_{\Omega'} \rho d\rho d\theta dz$

(球坐标) $V = \iiint_{\Omega'} r^2 \sin\varphi dr d\varphi d\theta$

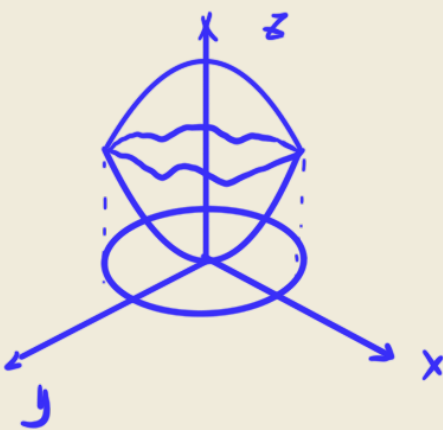
例 6.4.1 $z = x^2 + y^2$ 与 $z = 2 - \sqrt{x^2 + y^2}$ 所围的立体体积

$$\begin{aligned} V &= \int_0^1 dz \iint_D d\sigma + \int_1^2 dz \iint_{D'} d\sigma = \pi \int_0^1 z dz + \pi \int_1^2 (2-z)^2 dz \\ &= \frac{\pi}{2} + \pi \left(4z - 2z^2 + \frac{1}{3}z^3 \right) \Big|_1^2 \\ &= \frac{5\pi}{6} \end{aligned}$$

$$\Omega = \{(x, y, z) \mid x^2 + y^2 \leq z \leq 2 - \sqrt{x^2 + y^2}, x^2 + y^2 \leq 1\}$$

$$V = \iint_D (2 - \sqrt{x^2 + y^2} - (x^2 + y^2)) dx dy = \int_0^{2\pi} d\theta \int_0^1 (2 - \rho - \rho^2) \cdot \rho d\rho = \frac{5\pi}{6}$$

例: 求曲面 $z = b - x^2 - y^2$ 与 $z = x^2 + 3y^2$ 的立体体积 (投影法, 找交线, 先-后=)



$$b - x^2 - y^2 = x^2 + 3y^2 \Rightarrow x^2 + 2y^2 = 3 \text{ 交出椭圆}$$

$$D = \{(x, y) \mid x^2 + 2y^2 \leq 3\}$$

$$\Omega = \{(x, y, z) \mid (x, y) \in D, x^2 + 3y^2 \leq z \leq b - x^2 - y^2\}$$

$$V = \iint_D (b - x^2 - y^2 - (x^2 + 3y^2)) dx dy$$

$$\begin{cases} x = \sqrt{3} \rho \cos\theta \\ y = \frac{\sqrt{3}}{2} \rho \sin\theta \end{cases}$$

$$J(\rho, \theta) = \frac{3}{\sqrt{2}} \rho$$

$$\begin{aligned}
 \text{原式} &= \int_0^1 dp \int_0^{2\pi} (b - b\rho^2 \cos\theta - b\rho^2 \sin^2\theta) \frac{3}{\sqrt{2}} \rho d\theta \\
 &= b \cdot \frac{3}{\sqrt{2}} \cdot 2\pi \int_0^1 \rho(1-\rho^2) d\rho \\
 &= 18\sqrt{2}\pi \cdot \left(\frac{1}{2}\rho^2 - \frac{1}{4}\rho^4\right) \Big|_0^1 \\
 &= \frac{9\sqrt{2}}{2} \pi
 \end{aligned}$$

曲面面积

$$\lim_{\lambda \rightarrow 0} \sum_{i=1}^n \sigma(\pi_i)$$

$$\begin{aligned}
 \Delta S &= |(\Delta x, 0, z(x+\Delta x, y) - z(x, y)) \times (0, \Delta y, z(x, y+\Delta y) - z(x, y))| \\
 &\approx |(1, 0, \frac{\partial z}{\partial x}) \times (0, 1, \frac{\partial z}{\partial y}) \Delta x \Delta y|
 \end{aligned}$$

$$S = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dx dy$$

参数方程下曲面的面积公式

$$x = x(u, v) \quad y = y(u, v) \quad z = z(u, v)$$

$$A = \frac{D(y, z)}{D(u, v)} \quad B = \frac{D(z, x)}{D(u, v)} \quad C = \frac{D(x, y)}{D(u, v)}$$

$$S = \iint_D \sqrt{A^2 + B^2 + C^2} du dv$$

$$ds = \sqrt{A^2 + B^2 + C^2} du dv \quad \text{曲面微元}$$

$$ds = |(\vec{r}_u \cdot \Delta u) \times (\vec{r}_v \cdot \Delta v)| = |\vec{r}_u \times \vec{r}_v| du dv = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{vmatrix} = (A, B, C)$$

$$|\vec{a} \times \vec{b}| = \sqrt{|\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2}$$

$$\Rightarrow ds = \sqrt{|\vec{r}_u|^2 |\vec{r}_v|^2 - (\vec{r}_u \cdot \vec{r}_v)^2}$$

对于 $\vec{r}(x,y) = (x,y,z(x,y))$

直角坐标下曲面的面积公式, 设光滑曲面 S 的方程 $z = f(x,y)$

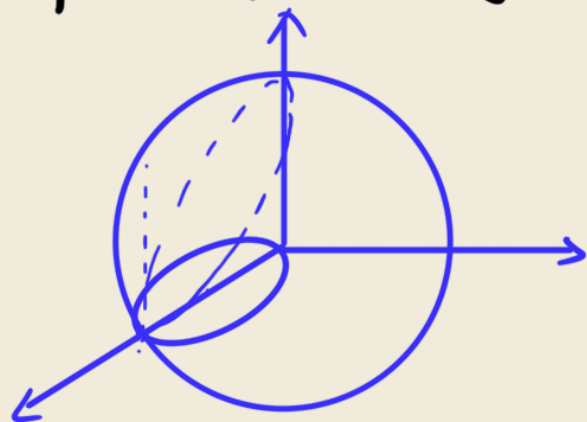
$$\text{有 } S = \iint_D \sqrt{1 + (f'_x)^2 + (f'_y)^2} dx dy$$

$$\text{证: } \vec{r}'_x = (1, 0, f'_x) \quad \vec{r}'_y = (0, 1, f'_y)$$

$$\vec{r}'_x \times \vec{r}'_y = (-f'_x, -f'_y, 1)$$

$$\Rightarrow S = \iint_D \sqrt{1 + (f'_x)^2 + (f'_y)^2}$$

例 6.4.3 求球面 $x^2 + y^2 + z^2 = a^2$ 被柱面 $x^2 + y^2 = ax$ 所截的部分曲面面积



$$(x - \frac{a}{2})^2 + y^2 = \frac{a^2}{4}$$

$$D \{ (x,y) \mid x^2 + y^2 \leq ax \} \quad \rho^2 \leq a \rho \cos \theta$$

$$\vec{r}'_x = (1, 0, \frac{-x}{\sqrt{a^2 - x^2 - y^2}}) \quad \vec{r}'_y = (0, 1, \frac{-y}{\sqrt{a^2 - x^2 - y^2}})$$

$$\vec{r}'_x \times \vec{r}'_y = (\frac{x}{\sqrt{a^2 - x^2 - y^2}}, \frac{y}{\sqrt{a^2 - x^2 - y^2}}, 1)$$

$$\begin{aligned} S &= 4 \iint_D \sqrt{\frac{x^2}{a^2 - x^2 - y^2} + \frac{y^2}{a^2 - x^2 - y^2} + 1} dx dy = 4 \iint_D \frac{a}{\sqrt{a^2 - x^2 - y^2}} dx dy \\ &= 4a \int_0^{2\pi} d\theta \int_0^{a \cos \theta} \frac{\rho}{\sqrt{a^2 - \rho^2}} d\rho \\ &= 2a^2(\pi - 2) \end{aligned}$$

1. 参数化曲面

$$\vec{r}(x, y) = (x, y, \sqrt{a^2 - x^2 - y^2})$$

$$\vec{r}(\rho, \theta) = \left(\frac{a}{2} + \rho \cos \theta, \rho \sin \theta, \sqrt{a^2 - \left(\frac{a}{2} + \rho \cos \theta\right)^2 - (\rho \sin \theta)^2}\right)$$

2. 确定参数范围

$$D = \{(x, y) \mid x^2 + y^2 \leq ax\}$$

$$D = \{(\theta, \rho) \mid 0 \leq \theta \leq 2\pi, 0 \leq \rho \leq \frac{a}{2}\}$$

例 6.4.4 求柱面 $x^2 + y^2 = ax$ 位于球面 $x^2 + y^2 + z^2 = a^2$ 内的部分曲面的面积

设 S_1 为第一卦限内面积

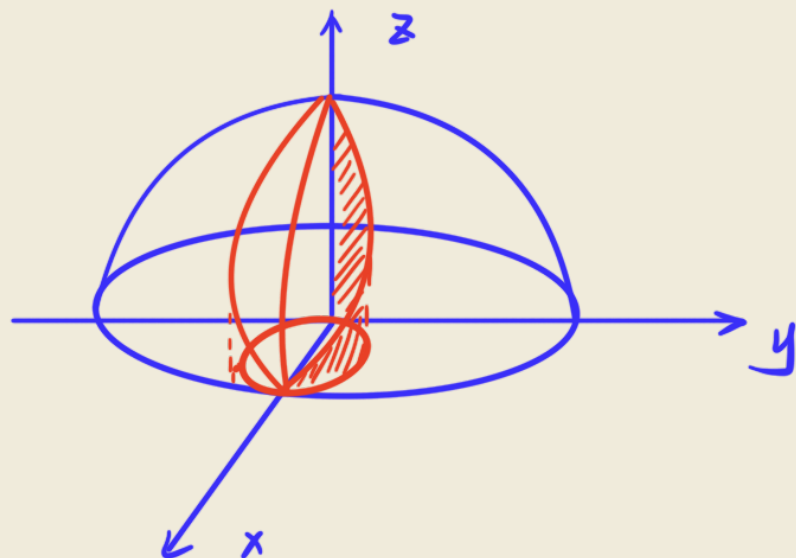
$$\begin{cases} x = \frac{a}{2} + \frac{a}{2} \cos \theta & \theta \in [0, 2\pi] \\ y = 0 + \frac{a}{2} \sin \theta & 0 \leq z \leq \sqrt{a^2 - \left(\frac{a}{2} + \frac{a}{2} \cos \theta\right)^2 - \left(\frac{a}{2} \sin \theta\right)^2} \\ z = z \end{cases}$$

$$ds = |\vec{r}_\theta \times \vec{r}_z| = \left| \left(-\frac{a}{2} \sin \theta, \frac{a}{2} \cos \theta, 0\right) \times (0, 0, 1) \right| = \left| \frac{a}{2} \cos \theta, \frac{a}{2} \sin \theta, 0 \right| = \frac{a}{2}$$

$$S = 2 \cdot \frac{a}{2} \cdot \int_0^{2\pi} d\theta \int_0^{\sqrt{\frac{a^2}{2}(1-\cos \theta)}} dz = 2 \cdot \frac{a^2}{2} \int_0^{2\pi} \left| \sin \frac{\theta}{2} \right| d\theta = 4a^2$$

$$\vec{r}(x, z) = (x, \sqrt{ax-x^2}, z)$$

$$z^2 = a^2 - ax \Rightarrow 0 \leq z \leq \sqrt{a^2 - ax}, 0 \leq x \leq a$$



设 D 是由心脏线 $\rho = a(1 + \cos\theta)$ 所围质量均匀的薄片, 求它对 y 轴的转动惯量

$$I_y = \iint_D x^2 k \, dx \, dy$$

$$I_y = k \int_0^{2\pi} d\theta \int_0^{a(1+\cos\theta)} \rho^3 \cos^2\theta \, d\rho = \frac{49}{32} k \pi a^4$$